

RESEARCH ARTICLE

Saturated Hydraulic Conductivity in a Southern Patagonian Peatland: Estimation Methods and Associated Properties

Mariano S. Iseas^{1,2}  | Hugo A. Pereyra¹ | Verónica A. Pancotto^{1,3}

¹Laboratorio de Ecología Terrestre y Acuática, Centro Austral de Investigaciones Científicas (CADIC), consejo Nacional de Investigaciones Científicas y Técnica (CONICET), Ushuaia, Argentina | ²Cátedra de Física, Departamento de Ingeniería Agrícola y Uso de la Tierra, Facultad de Agronomía, Universidad de Buenos Aires (FAUBA), Ciudad Autónoma de Buenos Aires, Argentina | ³Instituto de Ciencias Polares, Ambiente y Recursos Naturales, Universidad Nacional de Tierra del Fuego (ICPA-UNTDF), Ushuaia, Argentina

Correspondence: Mariano S. Iseas (miseas@agro.uba.ar)

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ABSTRACT

Saturated hydraulic conductivity (K) is often used to characterise the hydrological dynamics of peatlands. There are several methods to determine hydraulic conductivity, both in the field and in the laboratory. In the field, the piezometer method is one of the most widely used, while in the laboratory, the modified cube method (MCM) is often used in peaty soils due to its practicality. It also allows the estimation of anisotropy as K can be measured vertically or horizontally. Patagonia hosts the largest concentration of peatlands in South America, which are recognised for their pristine conservation status. However, there are few studies that address the hydrological dynamics of peatlands in the region. In this work, we aimed to investigate the relationships of K with other properties in southern peatlands and relate it to peat type classification. We also evaluate the performance of a modified MCM method, using a variable falling head; finally, we compare the laboratory and field methods used. The study was carried out in a pristine raised bog in southern Tierra del Fuego (Argentina). K was a function of depth, bulk density, humification index, and root density. Peat type classification adequately summarised these ecohydrological properties. Anisotropy was observed, but no preferential flow direction was evident. Both methods were consistent, showing that high and low permeability sites were well represented both in the field and in the laboratory. The alternative MCM method using the falling head mode obtained low experimental error and high correlation with the traditional MCM method. This method proved to be an interesting alternative as its application is simpler, reliable and recommended especially for low permeability samples.

1 | Introduction

Peatlands are important global carbon sinks (Yu et al. 2010). However, this capacity is directly and indirectly regulated by their hydrological dynamics. Hydrology controls many biogeochemical and ecological processes such as carbon cycling, nutrient dynamics, oxygen and gas diffusion and solubility, and plant composition and diversity (Holden 2006; Morris et al. 2011a). It also regulates interactions with downstream terrestrial and

aquatic ecosystems by facilitating the transfer of matter and energy through groundwater discharge and/or surface runoff (Strack et al. 2008; Waldron et al. 2009).

Saturated hydraulic conductivity (K) is an important attribute of peat that characterises hydrological dynamics and is often used in the development of ecohydrological models (Herbst et al. 2006; Morris et al. 2011a; Nijp, Metselaar, Limpens, Teutschbein, et al. 2017a). Important ecosystem services such as

water storage capacity, water regulation or contaminants filtering/buffering depend on this property (Lennartz and Liu 2019; Shuttleworth et al. 2015). In turn, it is conditioned by the degree of decomposition, depth, porosity, bulk density and vegetation composition of the peat (Crockett et al. 2016; Morris et al. 2022; Paat et al. 2022). It is also indirectly influenced by peatland type, microtopography, peat age, slope, and relative spatial position in the bog (margin or center) (Holden 2006; Lewis et al. 2012; Branham and Strack, 2014).

There are several methods for determining K, both in the laboratory and in the field. Most of them follow Darcy's principle of flow in a porous media, in which volumetric flow rate of water per unit cross-sectional area of the porous medium is directly proportional to the hydraulic gradient and a constant (K). Alternatively, in these methods, the hydraulic gradient can be performed as: (1) Constant head over test time, or (2) variable head, which could be rising head or falling head during the test time (Klute and Dirksen 1986). In the field, the most common method is the piezometric test, which consists of removing or adding a known volume of water (slug and bail) and evaluating the head recovery over time (Hogan et al. 2006; Surridge et al. 2005). On the other hand, permeameters are used in the laboratory to perform tests on undisturbed samples (Rycroft et al. 1975; Hoag and Price 1997). The "cube method" developed by Bouma and Dekker (1981) with later modifications (MCM) is often used on peat soils, where a cubic sample is sealed on its faces and a hydraulic gradient is applied with a constant head mode (Beckwith et al. 2003; Kruse et al. 2008; Rosa and Larocque 2008). An interesting feature of this method is that K can be determined in both the horizontal and vertical directions; this makes it possible to assess the anisotropy. The estimating of the 'true K' is not straightforward; different methods are accurate but not exact, making it difficult to compare (Surridge et al. 2005). However, it is important that comparisons between methods are made using the same hydraulic gradient mode (constant or variable head) as they refer to stationary or transient flows, as well as to the orientation of the flow during the test (horizontal or vertical). In general, the variable head mode is used for low permeability substrates and the constant head mode for high permeability substrates (Bagarello et al. 2004; Klute and Dirksen 1986).

The hydrology of Patagonian peatlands has been poorly examined (León et al. 2021), resulting in a lack of reference values for hydraulic parameters such as K. Differences in formation, climate, and vegetation between Patagonian and northern peatlands which have been extensively studied may result in distinct hydrogeological characteristics. To our knowledge, only two studies have monitored hydraulic conductivity in the region. Broder et al. (2012) conducted slug tests to estimate catotelm K at three depths in a cushion peatland near Punta Arenas, Chile. Meanwhile, Couwenberg et al. (2022) used a piezometric method to measure K at varying depths in an ombrotrophic peatland near Ushuaia, Tierra del Fuego. In this context, local hydrological research is needed to elucidate potential changes in biogeochemical processes of these peatlands under a climate change scenario.

The objectives of this work were:

- (i) To study, in the field and laboratory, the saturated hydraulic conductivity (K) and anisotropy of *Sphagnum* peatlands in southern Patagonia and its relationship with peat properties as organic matter, bulk density, root density, and degree of decomposition; (ii) In turn, to analyse if peat properties respond to a classification of peat type; (iii) To evaluate the effectiveness and reproducibility of an alternative MCM method using falling head mode to estimate K; and (iv) to compare laboratory and field methods for K determination.

2 | Methodology

2.1 | Sampling Site

Sampling and field tests were carried out in an undisturbed ombrotrophic rised bog located in the southern area of Tierra del Fuego (Argentina), (54°43'10.81"S; 68°4'12.13"W) (Figure 1b). The regional climate is temperate oceanic, with a mean annual temperature of 5.3°C and precipitation of 578 mm. The warmest conditions occur in January, with mean temperatures around 9°C, whereas July is the coldest month, averaging 1.5°C (Couwenberg et al. 2022). The site is representative of the *Sphagnum magellanicum* peat bogs, that extend along the valleys of the Fuegian Andes, in a mosaic with *Nothofagus* spp. forests. The microtopography is dominated by a complex of hummock-hollows and lawns, and the average peat depth varies between 3 and 4 m (León et al. 2021). Downstream of the bog, the beaver dam influences the flow that eventually reaches the Lasifashaj River (Figure 1a). The water level was monitored during the 2021–2022 growing season, with a piezometer installed in a hollow, which fluctuated between −10 and −28 cm below the surface. (Iseas et al. 2025). Vascular vegetation grows on a matrix of living or senescent moss, covering on average 46% of the surface (Figure 1c). It is mainly composed of perennial herbs (*Marsippospermum grandiflorum*, *Tetroncium magellanicum*, *Nanodea muscosa*, *Carex magellanica*), shrubs (*Empetrum rubrum*, *Gaultheria pumila*, *Chilotrimum diffusum*), and dwarf trees (*Nothofagus antarctica* and *N. betuloides*).

2.2 | Peat Sampling and Laboratory Measurements

Four sampling sites were selected, two in the lawn area (points A and B) and two in the hummock-hollow complex (points C and D), as illustrated in Figure 1b. The field tests and sampling were performed in April 2022 (autumn), when the water level was high. A steel peat monolith sampler, commercialised by Eijkelkamp Soil and Water (Giesbeek, the Netherlands) and developed by Wardenaar (1987), was utilised to obtain cores with dimensions of approximately 10×10×100 cm. The live *Sphagnum* was previously removed from 7 to 10 cm, and finally the maximum sampling depth varied between 60 and 75 cm at each point due to loss of structure of the last few centimetres during extraction. The cores (Figure 1d) were carefully placed in slim rectangular metal containers to ensure structural integrity during transportation, and then promptly froze until fractionation. In the laboratory we cut homogeneous 6 cm thick peat samples from the core with a sharp serrated knife at different depths until reaching the maximum. As a result, a total of 26

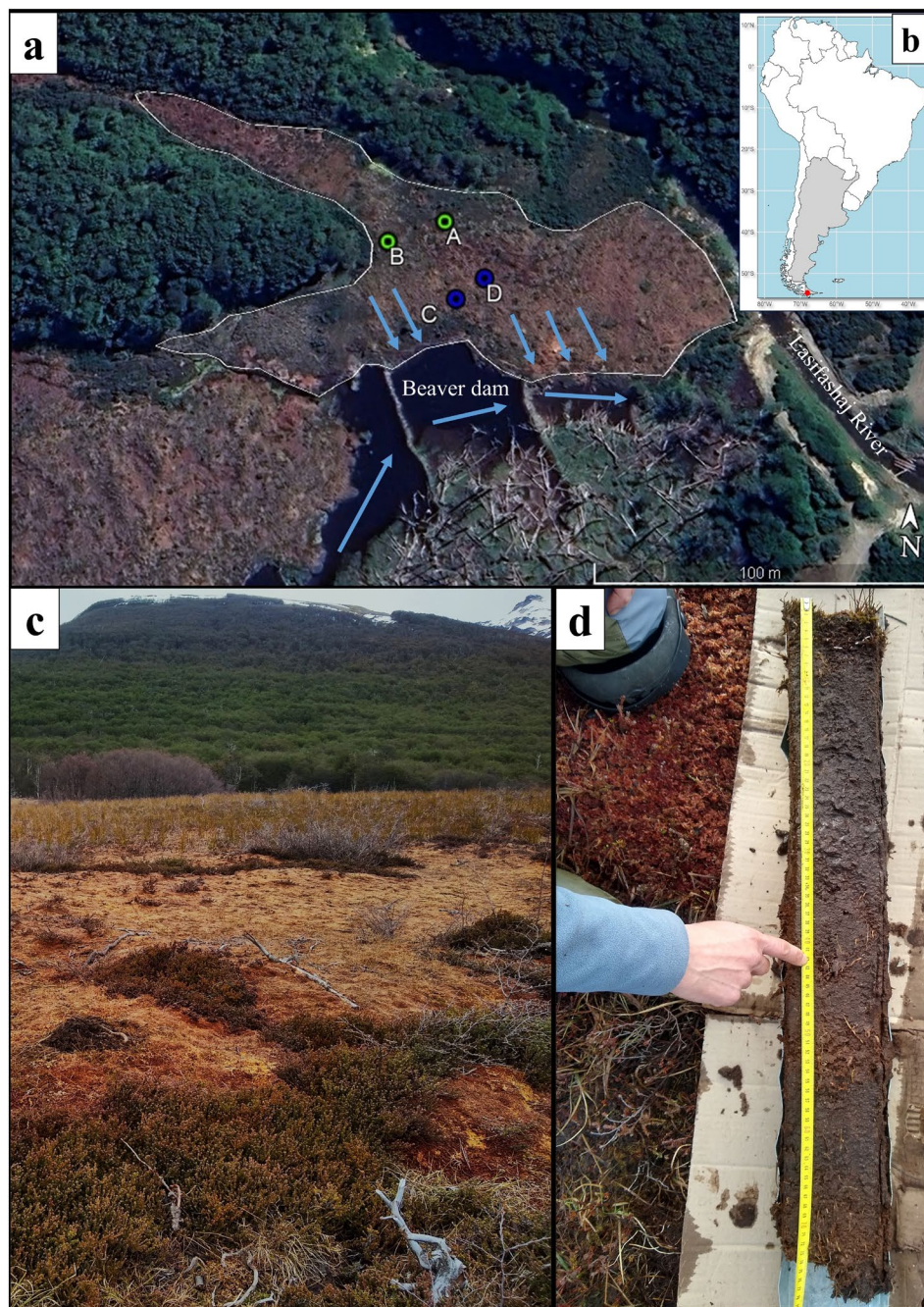


FIGURE 1 | (a) Satellite picture of the study area with the location of the sampling sites. Green points indicate lawn microforms (A and B), and blue points indicate hummock microforms (C and D). Blue arrows indicate direction of water flow; (b) South America map with site location (red point); (c) Studied bog in a landscape view, with the complex hollow- hummock in the foreground; (d) Core peat profile extracted from the lawn.

samples ($L = 6$ cm) of undisturbed peat were obtained at varying depths. In A, B and D, 6 layers were obtained and 8 in C.

Finally, a visual classification was conducted based on colour, plant composition, and root abundance, following the approach proposed by Paat et al. (2022). The classification criteria categorised light-coloured peat with undecomposed *Sphagnum* as Ombrotrophic, dark layers rich in highly decomposed vascular plant remains as Minerotrophic, and mixed layers with characteristics of both as Transitional. Accordingly, each layer was classified as Ombrotrophic, Transitional or Minerotrophic (Figure 2A).

2.3 | K Laboratory Determination

From the selected layers, inner cubes of 6 cm per side (216 cm^3) were trimmed (Figure 2A). This procedure ensured that the peat cube samples were free from deformation or structural disturbance associated with the core edges during extraction and transport. The lateral and bottom faces of the peat cube were coated with a layer of paraffin using a paper mould. Then, the top face—which had a graduated plastic pipette (2.5 cm in diameter) attached—was sealed onto the cube's lateral edges, leaving a small headspace above the sample (Figure 2B). Finally, a heated metal punch was used to open the bottom

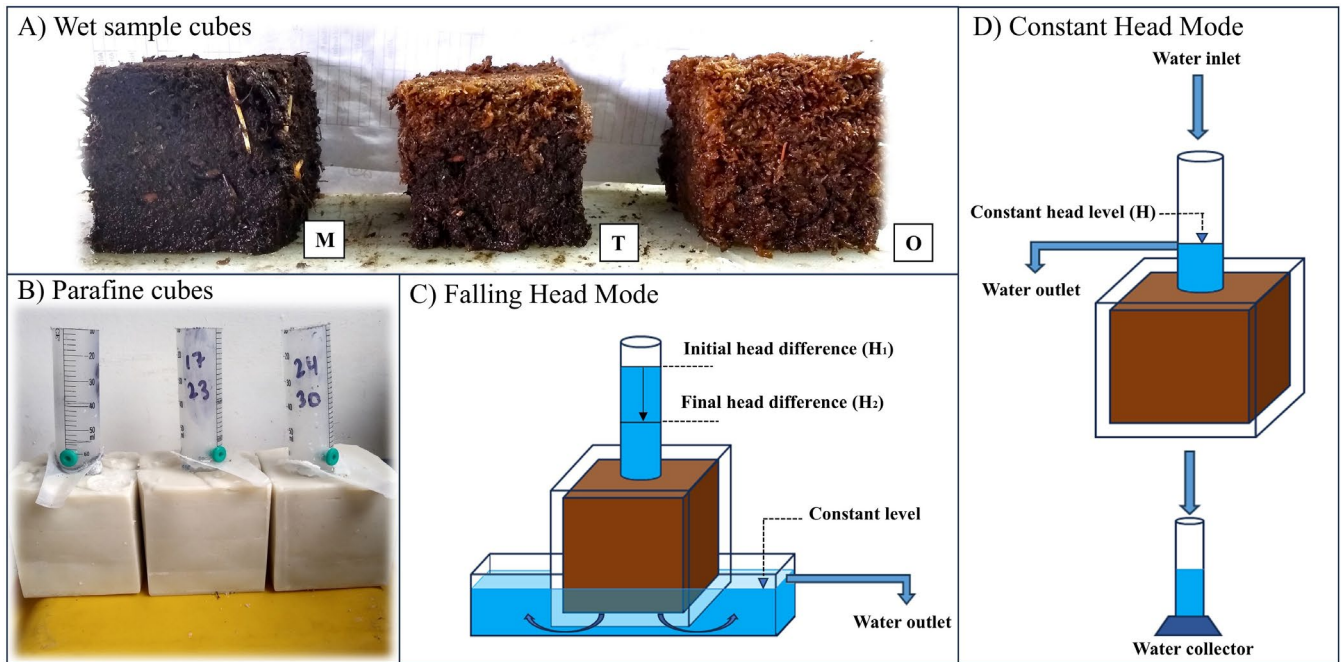


FIGURE 2 | (A) Saturated peat cubes, prior sealing with paraffin (O: Ombrotrophic peat, T: Transitional peat, M: Minerotrophic peat); (B) Image of the peat cube sealed with paraffin; (C) Schematic of the Falling head method used in the modified cube method; (D) Schematic of the Constant head method used in the modified cube method (H =hydraulic head difference). *Note:* For a more detailed description of the cube method, see the step-by-step diagram in Figure S1 in the [Supporting Information](#) section.

part, to create a drainage outlet (see detailed procedure in Figure S1). K was determined in these prepared cubes using two hydraulic head methods: a constant head approach, following the modified cube method (MCM) based on Beckwith et al. (2003) and the recommendations of Nijp et al. (2017b), and an alternative falling head approach.

In the falling head method, a variable hydraulic head (falling) is applied, ranging from an initial head $H_1 = 9.0$ cm to a final head $H_2 = 6.5$ cm, as shown in Figure 2C. These reference heights were previously marked on the graduated tube placed above the sample. In the test, the time required for the water level to drop from H_1 to H_2 is recorded, and K is calculated using Equation (1) following (Klute and Dirksen 1986).

$$K = \frac{a \times L}{t \times A} \times \ln \frac{H_1}{H_2} \quad (1)$$

where (a) is the tube area, (L) sample length, and (A) cross-sectional area of the peat cube.

The constant head method applies a constant hydraulic head difference (H) of 8 cm to the sample (2 cm of water column above the sample plus 6 cm corresponding to the saturated peat sample) and measuring the time (t) at which 50 mL (V) is collected in a graduated cylinder at the outlet. The outflow was allowed to stabilise before starting the measurements. Constant head conditions were maintained by supplying water from the top of the tube at a flow rate much higher than the sample drainage, while the outflow was controlled through a perforation in the tube at the constant head level (Figure 2D). Equation (2) is used to calculate K by the constant head method.

$$K = \frac{V \times L}{t \times A \times H} \quad (2)$$

where (A) is the cross-sectional area of the peat cube and (L) is the sample length.

K determinations were estimated in both vertical (KV) and horizontal (KH) orientations for each peat cube. For this purpose, once the vertical orientation was determined, the top face—which contained the tube—was carefully removed using a heated knife and reattached to one of the lateral faces, leaving the bottom face open to allow flow discharge (see Figure S1). All the samples were measured in the following order: constant head mode vertical orientation, falling head mode vertical orientation, constant head mode horizontal orientation, and falling head mode horizontal orientation. Each determination was repeated 3 times. All measurements were standardised to 20°C (Klute and Dirksen 1986), and previously the sample cubes were saturated with distilled water for 12 hs. The anisotropy for both laboratory methods of each sample was calculated as described by Branham and Strack (2014) from the K determined in the vertical and horizontal directions, defined by the following equation.

$$Anisotropy = \log_{10} \left(\frac{KH}{KV} \right) \quad (3)$$

Positive values indicate preferential flow in the horizontal direction, while negative values indicate preferential flow in the vertical direction. Anisotropy, either vertical or horizontal, is considered to occur when the relative difference between KH and KV exceeds 20% (Branham and Strack 2014).

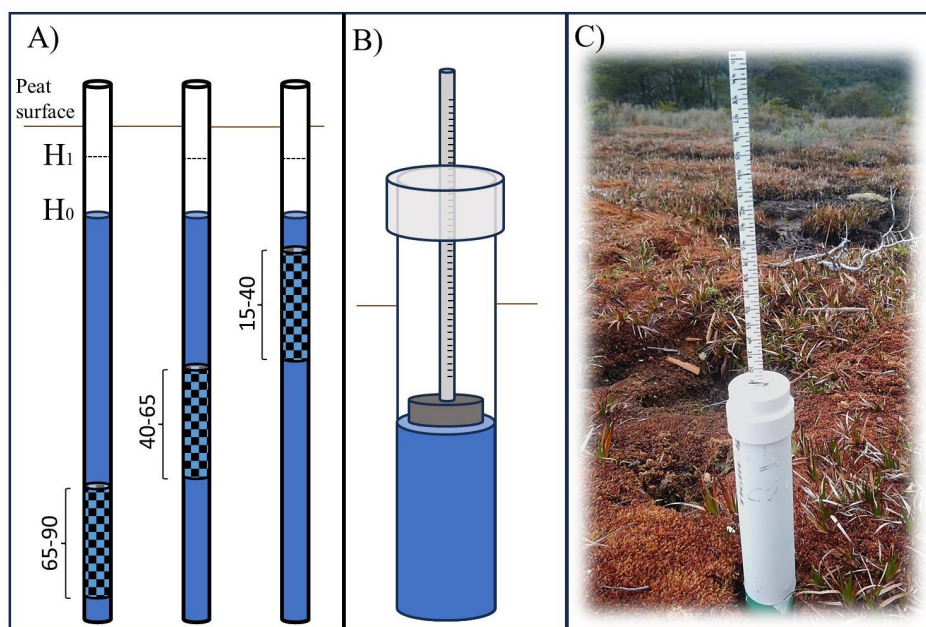


FIGURE 3 | (a) Sampling design of piezometer tests with depth ranges (cm) (H_0 = water level, H_1 = initial head difference); (b) illustration of the float used to monitor head recovery; (c) photo of a piezometer during the field test.

2.4 | K Field Determination

We used the piezometer test method to determine field K (K_{field}) following Freeze and Cherry (1979) and the recommendations of Surridge et al. (2005). At the same sites where samples were taken for the laboratory determinations (A, B, C and D), PVC piezometers ($D_{\text{int}} = 3.5$ cm) were installed at three ranges of depths (cm): 15–40, 40–65 and 65–90 (Figure 3). A 25 cm length of the piezometers was perforated at a ratio of > 65% (of inlet open to water flow) at each depth and covered with a 0.5 mm mesh to prevent the entry of coarse particles. Slug tests were conducted with a variable head of approximately 10 cm. “Pumping tests” were used when the water level was close to the surface, and “loading tests” when it was close to the intake zone. The piezometric head was monitored with a floating graduated rod (Figure 3B) and recorded over time until at least 70% recovery was reached. A minimum of two tests were performed at each site and depth. We use the Hvorslev (1951) equation (4) to obtain the K defined as:

$$K = \frac{A}{Ft} \log \frac{H_0}{H_1} \quad (4)$$

where H_0 is the initial head difference between the piezometric level (or water level) and peat surface (cm), H_1 is the head difference at the end of the test (t) or when the recovery was 70%, A is the inside cross-sectional area of the piezometer standpipe (cm^2) and F is a shape factor for closed-bottom cylindrical pipes calculated from Brand and Premchitt (1980, 1982).

2.5 | Peat Properties

Bulk density (BD, g cm^{-3}), organic matter (OM, %), root density (RD, mg g^{-1} dry peat weight), and humification index (HUM) were determined on each peat cube (after K determination)

following Chambers et al. (2011). To determine the volume, the peat cubes were carefully covered with plastic film (to avoid destruction) and completely immersed in distilled water in a graduated container (cm^3). They were dried in an oven at 105°C to determine the BD. Then the coarse roots (> 2 mm) were manually separated from the peat to calculate RD. The samples were then ground with a coffee grinder to homogenise the material. For OM determination, 1 g of the dried and crushed peat samples were weighed into crucibles and muffled at 550°C for 4 h. Finally, HUM determination was performed using a colorimetric method, following the protocol described by Chambers et al. (2011). This method involves the extraction of humic acids using a NaOH solution and their quantification based on absorbance at 540 nm measured with a spectrophotometer. Although the Von Post method is widely used to qualitatively assess the degree of peat decomposition, the colorimetric method was selected due to its simplicity, objectivity, and quantitative measurement (Paat et al. 2022).

2.6 | Data Analysis

Due to the high variability of K, its values were log 10 transformed to meet the assumptions of normality and homoscedasticity. As the studied thickness was greater in the field K determination (25 cm) compared to the smaller thickness in the laboratory K measurements (6 cm), laboratory values were averaged to match the corresponding field layer thickness for comparison.

Data analysis was performed using a linear mixed model (LME) from the *lme4* package, following the approach suggested by Morris et al. (2015, 2019), due to the lack of independence among samples from layers of the same core. First, the evaluation of the categories classified from plant remains (Ombrotrophic, Transitional and Minerotrophic) was conducted using ANOVA and contrast analysis (using “emmeans” package) for each

TABLE 1 | Mean saturated hydraulic conductivity (K , cm day^{-1}) measured in the field and laboratory, by site and depth. Depth is reported as the centred depth for the laboratory method and as depth intervals for the field method. For comparison, laboratory K values were averaged within the depth intervals corresponding to each field measurement.

Site	Field depth (cm)	Lab centred depth (cm)	N_{lab}	K_{Field}	KH_{CH}	KH_{FH}	KV_{CH}	KV_{FH}
				cm day^{-1}				
A	—	10	1	—	5.2×10^2	4.8×10^2	3.6×10^2	2.8×10^2
	15–40	25 35	2	7.8×10^1	2.5×10^2	2.1×10^2	1.4×10^2	1.1×10^2
	40–65	42 51	2	4.0×10^{-2}	2.4×10^2	2.1×10^2	1.7×10^2	1.1×10^2
	65–90	66	1	7.0×10^{-2}	1.3×10^2	8.3×10^1	1.8×10^2	1.3×10^2
B	—	10	1	—	1.2×10^2	1.0×10^2	2.2×10^2	2.0×10^2
	15–40	25 40	2	4.3×10^1	2.6×10^2	1.9×10^2	4.4×10^2	3.4×10^2
	40–65	50 60	2	6.0×10^{-1}	3.4×10^2	3.0×10^2	9.6×10^1	6.4×10^1
	65–90	69	1	5.8×10^{-1}	1.7×10^2	1.7×10^2	1.7×10^2	1.2×10^2
C	—	11	1	—	1.5×10^3	1.0×10^3	2.4×10^3	2.6×10^3
	15–40	18 25 33	3	3.0×10^2	7.4×10^2	4.1×10^2	1.0×10^3	8.7×10^2
	40–65	42 50 60	3	1.1×10^{-1}	1.1×10^2	5.1×10^1	9.1×10^1	4.8×10^1
	65–90	66	1	4.0×10^{-2}	2.9×10^1	1.0×10^1	7.2×10^1	3.7×10^1
D	—	13	1	—	3.1×10^3	2.4×10^3	3.8×10^3	3.4×10^3
	15–40	20 27 35	3	2.9×10^2	1.7×10^2	1.5×10^2	6.6×10^2	5.2×10^2
	40–65	43 57	2	1.4×10^1	2.0×10^2	1.8×10^2	1.4×10^2	9.1×10^1
	65–90	—	—	2.4×10^{-1}	—	—	—	—

Abbreviations: K_{Field} = saturated hydraulic conductivity measured in the field over a 25 cm thick layer; KH_{CH} = horizontal saturated hydraulic conductivity at laboratory with constant head mode; KH_{FH} and KV_{FH} indicate horizontal and vertical saturated hydraulic conductivity at laboratory with falling head mode, respectively; KV_{CH} = vertical saturated hydraulic conductivity at laboratory with constant head mode; N_{lab} = number of laboratory samples (6 cm thickness each) averaged to represent the corresponding field depth interval.

variable studied (separately). An LME was adjusted where the fixed effects were the categories, and the random effects were the sampling 'site' (peat profile). Then, Pearson correlation (PC) analyses were performed to compare the K values obtained from different methods and to examine the relationships with the studied variables. Finally, a multiple regression LME was fitted to the horizontal and vertical $\text{Log}_{10}(K)$ values determined in the laboratory with the constant head method, using the continuous variables corresponding to peat properties and peat type categories (fixed effects), while considering the random effect of the 'site'. The Akaike Information Criterion (AIC) was used to select the best model, analysing the models step by step from the most complex to the simplest. All statistical analysis and modelling were performed using R software (R Core Team 2020). A p -value of 0.05 was used for all analyses.

3 | Results

3.1 | Laboratory and Field Hydraulic Conductivity

K field ranged over four orders of magnitude, from 3.0×10^2 to $4.0 \times 10^{-2} \text{ cm day}^{-1}$. The arithmetic mean across all samples was 60.6 cm day^{-1} , whereas the geometric mean was substantially lower (2.0 cm day^{-1}), which is commonly used to represent effective permeability. Mean hydraulic conductivity was higher in hummocks

(120 cm day^{-1}) than in lawns (21 cm day^{-1}). Depth accounted for most of the observed variability ($F = 23.3$, $p < 0.01$), with the surface layer exhibiting the highest K values (178 cm day^{-1}).

Laboratory and field K showed the same trend, but K laboratory was always higher and less variable with depth (Table 1). The site D, on the hummock-hollow complex, had the highest K values (throughout the profile), while site A, on lawn microform, had the lowest values (lower conductivity). These results were consistent both in laboratory and field methods. In this sense, the correlation of the $\text{Log}_{10}(K)$ between field and laboratory determinations was acceptable in both orientations (Figure 4A), with Pearson coefficients (PC) between 0.57 and 0.77. However, laboratory determinations generally overestimated field measurements, with larger discrepancies at sites containing impermeable layers. Overestimation was smaller for the falling head method, and the strongest correlations were observed between vertical orientations in field and laboratory measurements. In addition, correlations between falling and constant head tests were high and statistically significant for both orientations (Figure 4B), with systematically lower K values for the falling head test (Table 1). The coefficients of variation (CV, %), calculated from repeated measurements for each determination, were lower in the constant head tests, indicating reduced experimental errors for this laboratory method. Specifically, KH exhibited a mean CV of 9.2% in the constant

head method compared to 17.8% in the falling head method, whereas KV showed 6.2% and 10.7%, respectively.

3.2 | Hydraulic Conductivity and Peat Properties

The peat samples analysed were obtained from field cores, with sampling depths ranging from 7 to 72 cm. No consistent sequence of layers was observed across the peat profiles. The horizons are interspersed with a 'zebra' pattern where the different peat types occur at any depth. BD was positively related to humification and negatively to OM, but not significantly related to depth or root density (Table 2). OM content was high in all samples, showed low variability (CV = 3.5%), and increased with depth while decreasing with BD. In contrast, root density showed high variability (CV = 85%) throughout the profile. Roots density was negatively associated with depth, although this trend was not significant ($p = 0.06$; Table 2). Finally, the humification index ranged from 0.12 to 0.87 nm and was independent from depth.

Peat properties showed significant relationships with the constant head test in both orientations (Figure 5). Layers with greater depth, higher humification index, and bulk density exhibited lower K values (Figure 5A–C, respectively). Pearson correlation coefficients (PC) between these properties and K were significant, ranging from -0.37 to -0.65 . In contrast, root density showed a positive correlation with K (Figure 5D), which was significant only in the vertical orientation (PC = 0.59, $p < 0.05$). Relationships obtained with the falling head method ($\text{Log}_{10}K$) followed the same trends (data not shown).

KH, humification index, and BD varied significantly with peat type ($p < 0.05$; Table 3). Strikingly, KV did not differ significantly, although it followed the same trend but with greater variability than in the horizontal orientation. In contrast, root density, OM, and depth did not vary with peat type. In conclusion, minerotrophic layers can be distinguished from ombrotrophic layers by their lower horizontal permeability (KH), higher BD, and higher humification index. Transitional

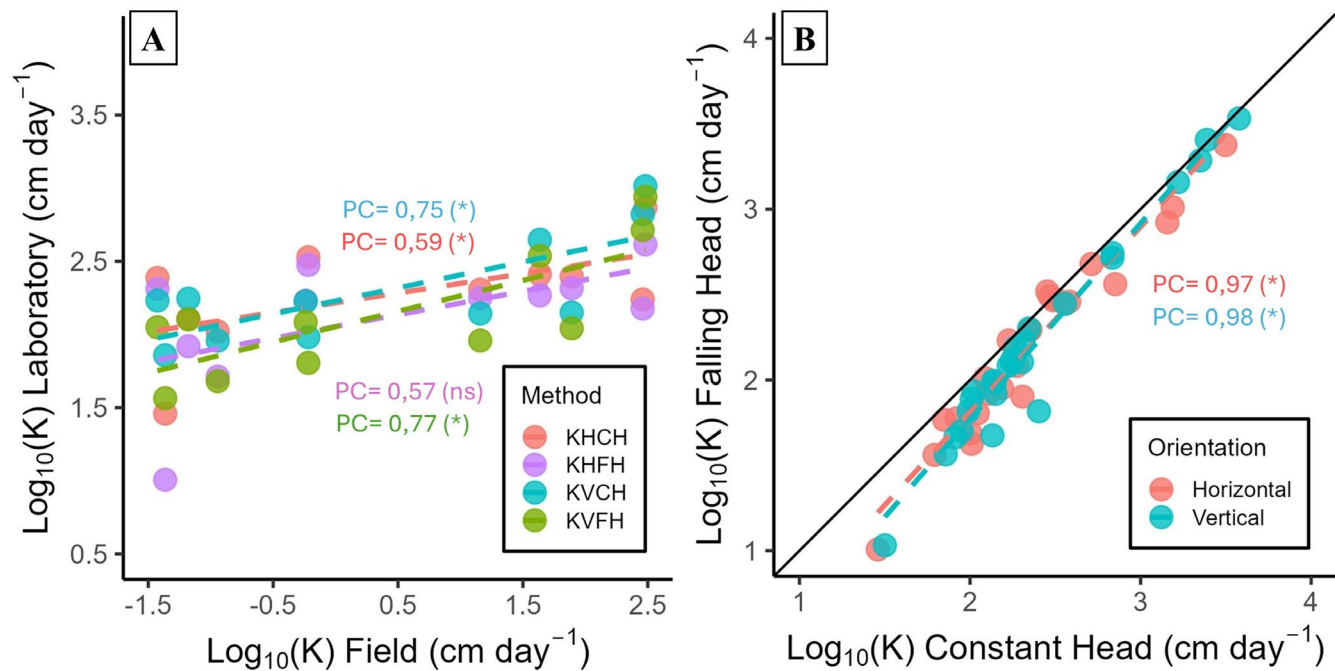


FIGURE 4 | (A) Correlation plot of the $\text{Log}_{10}(K)$ values between field (piezometric test) and laboratory methods (FH: Falling head and CH: Constant head) for horizontal and vertical orientations. (B) Correlation plot of the $\text{Log}_{10}(K)$ values obtained using laboratory constant head and falling head methods in both horizontal and vertical orientations. PC = Pearson coefficient, (*) significant correlation, (ns) no significant correlation.

TABLE 2 | Mean values and Pearson's correlation coefficients for the measured peat properties.

	Depth	BD	RD	OM	HUM	Mean + SD
Depth (cm)	1					36.8 ± 18.4
BD (g cm^{-3})	ns	1				0.09 ± 0.04
RD (g g DW^{-1})	-0.35	ns	1			67.7 ± 56.5
OM (%)	0.46*	-0.50*	ns	1		95.3 ± 3.3
HUM (nm)	ns	0.55*	ns	ns	1	0.36 ± 0.16

Note: Shaded cells above the diagonal indicate omitted redundant values, while the interaction coefficients are shown below the diagonal of the matrix.

Abbreviations: Depth, BD: Bulk density; HUM: Humification Index; Mean: arithmetic mean; ns: not significant; OM: Organic Matter Content; RD: Roots Density; SD: standard deviation.

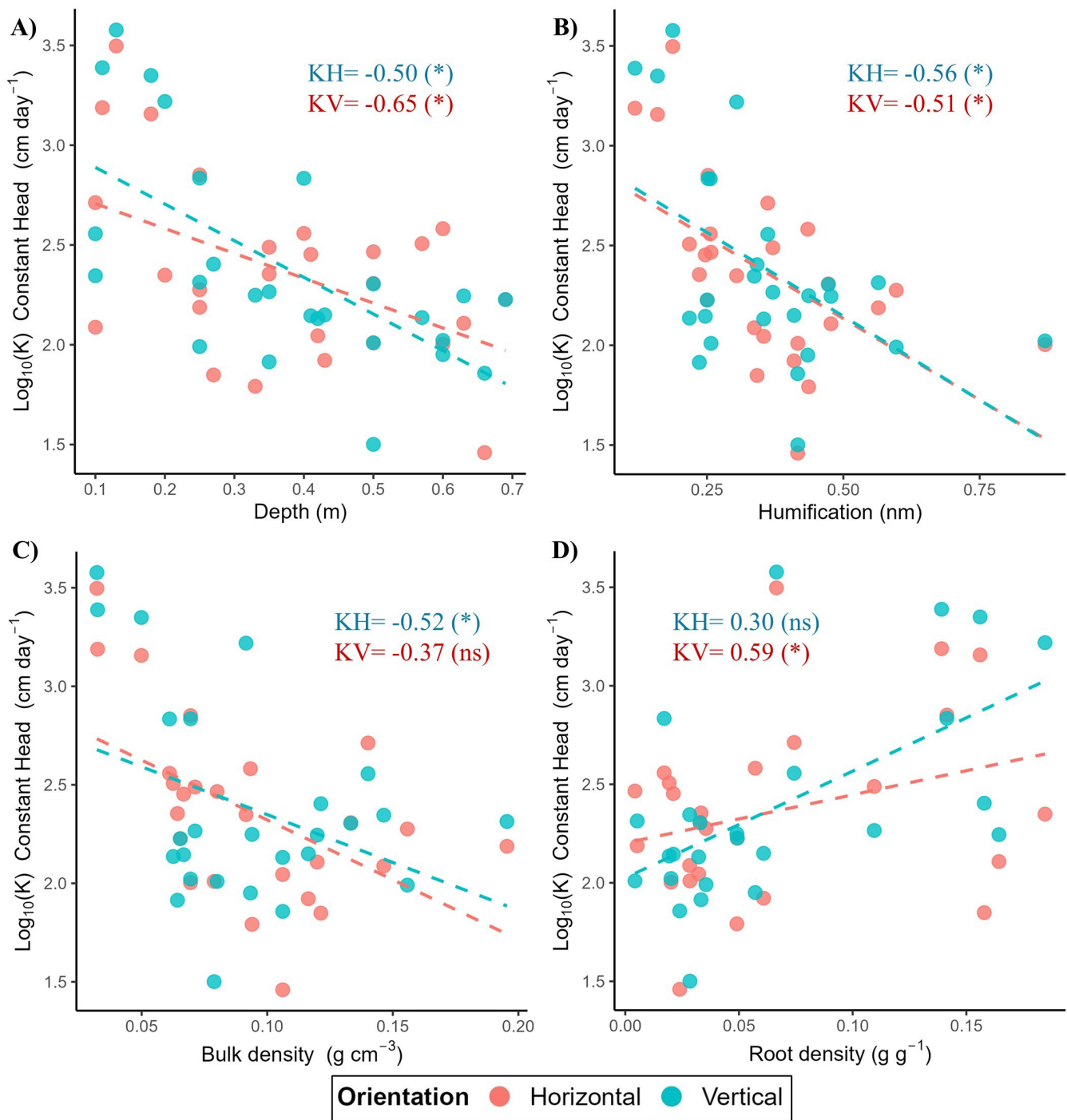


FIGURE 5 | Correlation plots of $\text{Log}_{10}(\text{K})$ values obtained in laboratory using the constant head method in both orientations versus (A) depth (cm), (B) humification index (nm), (C) bulk density (g cm^{-3}), and (D) root density (g g^{-1} DW). Values within each panel indicate the Pearson correlation coefficients for vertical (KV) and horizontal (KH) hydraulic conductivities. (*) indicates significant correlations at $p < 0.05$; (ns) denotes non-significant correlations.

layers generally showed values closer to minerotrophic layers for these properties.

3.3 | Anisotropy

Most of the samples (20 out of 26) exhibited anisotropy. Although individual values were positive or negative depending on whether $\text{KH} > \text{KV}$ or $\text{KV} > \text{KH}$, the mean absolute

percentage change in K between orientations was 0.88, indicating that hydraulic conductivity differed on average by nearly 90% between horizontal and vertical directions. Furthermore, these differences were significantly greater than 20% ($t = 3.40$; $p < 0.05$).

Despite the evident anisotropy among samples, there was no preferential trend towards either flow direction. The mean anisotropy value oscillated around zero, with a 95% confidence interval

ranging from -0.51 to 0.56 . The frequency of samples showing horizontal and vertical anisotropy was balanced, with 9 and 11 cases, respectively. Peat type classification showed no significant differences in this property (Table 3), and the same tendency was observed for both constant head and falling head methods.

Finally, in agreement with Figure 5C, a positive correlation was found between anisotropy values and root density ($PC=0.47$; $p<0.01$), indicating that layers with higher root abundance tended to exhibit preferentially vertical flow (not shown).

3.4 | Laboratory K Regression Model

The multiple linear regression model for vertical hydraulic conductivity determined with constant head, expressed as $\text{Log}_{10}(\text{KV})$, showed a negative relation with depth and positive with root density (Table 4). The categorical effect of the ombrotrophic layer type is a significant explanatory variable, positively related. The model explained 70% of the total variability in the response variable, with 68% accounted for by the fixed effects alone.

TABLE 3 | Results of the contrast analysis of the LME model with arithmetic mean and standard deviation (parentheses) values of the properties studied in each layer according to the proposed classification by peat type.

Variable	Ombrotrophic $n=10$	Transitional $n=9$	Minerotrophic $n=7$	p -value
KH_{CH} (cm day^{-1})	850 (190)	170 (200)	170 (230)	0.037
KV_{CH} (cm day^{-1})	1040 (270)	340 (290)	150 (330)	0.094
KH_{FH} (cm min^{-1})	620 (140)	140 (140)	110 (160)	0.035
KV_{FH} (cm day^{-1})	940 (260)	260 (270)	110 (310)	0.091
ANT_{CH}	-0.08 (0.11)	0.17 (0.12)	-0.03 (0.13)	0.298
ANT_{FH}	-0.02 (0.02)	0.03 (0.02)	-0.01 (0.03)	0.284
RD (mg g^{-1})	64.7 (58.6)	77.8 (60.2)	52.0 (52.4)	0.607
BD (g cm^{-3})	0.06 (0.02)	0.11 (0.03)	0.12 (0.04)	<0.001
OM (%)	96.3 (1.9)	93.9 (5.0)	95.7 (1.4)	0.121
HUM (nm)	0.22 (0.05)	0.37 (0.04)	0.55 (0.15)	<0.001
Depth (cm)	35.9 (19.3)	33.7 (18.8)	45.1 (17.0)	0.484

Abbreviations: ANT_{CH} : Anisotropy determined with constant head; ANT_{FH} : Anisotropy determined with falling head; BD: Bulk density; HUM: Humification index; KH_{CH} : Horizontal hydraulic conductivity with constant head; KH_{FH} : Horizontal hydraulic conductivity with falling head; KV_{CH} : Vertical hydraulic conductivity with constant head; KV_{FH} : Vertical hydraulic conductivity with falling head; OM: Organic matter; RD: Roots density.

TABLE 4 | Summary of Multiple Linear Regression Model to Describe $\text{Log}_{10}(\text{KV})$ of the vertical orientation determined with the constant head (CH) method.

Descriptor	Category	Coefficient	Standard error	t value	p -value
Constant	—	2.43	0.19	12.35	<0.001
Depth (m)	—	-1.40	0.35	-3.97	<0.001
RD (g g^{-1})	—	4.04	1.14	3.55	0.002
Layer Type	Ombrotrophic	0.45	0.14	3.20	0.003
	Transitional	—	—	—	—
	Minerotrophic	0.15	0.15	0.95	0.381

Note: Conditional $R^2=0.70$, Marginal $R^2=0.68$.

Following Equation (5) represents the obtained regression model:

$$\text{Log}_{10}(\text{KV}_{\text{CH}}) = 2.43 + 4.04 \text{ RD} - 1.40 \text{ Depth} + 0.45 \text{ Ombrotrophic} \quad (5)$$

In contrast, the regression model for $\text{Log}_{10}(\text{KH})$ is negatively related to depth and BD, and positively to the categorical ombrotrophic layers Equation (6) (Table 5). In this case, the full model explained most of the variability (71%) and the fixed effects alone explained 68%.

$$\text{Log}_{10}(\text{KH}_{\text{CH}}) = 3.15 - 5.07 \text{ BD} - 1.44 \text{ Depth} + 0.40 \text{ Ombrotrophic} \quad (6)$$

4 | Discussion

4.1 | Hydraulic Conductivity of Patagonian Peatlands

The high spatial variability of hydraulic conductivity measured in this study (ranging from 4.0×10^{-2} to $3.8 \times 10^3 \text{ cm day}^{-1}$) aligns with the wide range reported by Morris et al. (2022) based on both field and laboratory determinations. This

TABLE 5 | Summary of Multiple Linear Regression Model to Describe Log_{10} (KH) of the horizontal orientation determined with the constant head (CH) method.

Descriptor	Category	Coefficient	Standard error	t value	p-value
Constant	—	3.15	0.30	10.58	< 0.001
BD (g cm^{-3})	—	-5.07	2.12	-2.39	0.027
Depth (m)	—	-1.44	0.31	-4.72	< 0.001
Layer Type	<i>Ombrotrophic</i>	0.40	0.16	2.48	0.022
	<i>Transitional</i>	—	—	—	—
	<i>Minerotrophic</i>	0.29	0.14	2.04	0.055

Note: Conditional $R^2 = 0.71$, Marginal $R^2 = 0.68$.

property remains poorly characterised in Patagonian peatlands, with few field determinations available. Using piezometers, Couwenberg et al. (2022) reported K values of 8.1×10^2 and $1.2 \times 10^2 \text{ cm day}^{-1}$ for wet and dry sites, respectively, measured 15 cm below the water level in a pristine Fuegian bog. Similarly, Broder et al. (2012) reported a mean K of $5.2 \times 10^1 \text{ cm day}^{-1}$ at 100 cm depth in the catotelm near Punta Arenas, Chile. These values are comparable to our mean K field ($1.8 \times 10^2 \text{ cm day}^{-1}$) at the shallowest depth, also obtained with the piezometric method.

In agreement with Morris et al. (2022), the peat properties most strongly related to permeability were depth, degree of humification, bulk density (BD), and root density. Although hydraulic conductivity (K) decreased significantly with depth, this relationship should not be interpreted as linear. Instead, depth mainly reflects differences between the acrotelm and the catotelm peat layers. In the acrotelm, water levels fluctuate, compression is lower, and peat contains living vegetation, whereas the catotelm is permanently water-saturated, more compressed, and composed of dead, more humified plant material (Morris et al. 2015). Under these conditions, the acrotelm shows higher hydraulic conductivity and diffusivity, while water movement in the catotelm tends to be slower (Ingram 1978). Consequently, the relationship between depth and K was not consistent across the peat profile. Deeper layers showed K values similar to those near the surface. Accordingly, field K was similar within the catotelm (40–65 and 65–90 cm intervals), but increased significantly in the 15–40 cm interval corresponding to the acrotelm (Table 1). This suggests that the observed depth effect is mainly driven by contrasts between peat layers rather than by depth itself. Similarly, laboratory measurements using the MCM method within the catotelm (below 0.3 m) did not show a clear trend with depth (Figure 5A), although the most permeable samples were consistently located within the acrotelm. Comparable results have been reported in other peatlands, where no clear relationship between K and depth was found below the acrotelm (Holden and Burt 2003; Rosa and Larocque 2008). In these cases, permeability was instead controlled by local heterogeneity and peat layering. Therefore, our results show that hydraulic conductivity does not always decrease with depth, and that the catotelm is not uniformly impermeable, as is often assumed (Clymo 1984; Holden 2006). Also, recent studies

have shown that some catotelm layers can retain relatively high permeability (Baird et al. 2016; Morris, Waddington, et al. 2011b; Paat et al. 2024). In our study, deep layers with high K values were associated with buried layers of poorly decomposed Sphagnum.

On the other hand, we found that K decreases with increasing BD and humification index, and that in turn the latter is correlated with each other. These relationships have been reported by other authors using similar methodologies (Fewster et al. 2023; Morris et al. 2019; Paat et al. 2022). Taking this into account, it can be concluded that during the decomposition process of organic matter (humification), peat loses its original elastic and plastic structure, thus reducing its porosity (Quinton et al. 2008; Rezanezhad et al. 2016). This loss of macroporosity translates into an increase in BD and consequently less water conduction (K). In turn, the deeper, more decomposed layers suffer greater compression due to the weight of the upper layers, which also leads to an increase in BD (Liu and Lennartz 2019).

When comparing the estimated vertical and horizontal K, a certain degree of anisotropy was observed, although no systematic preferential flow direction (horizontal or vertical) was evident. Contrary to our results, many studies have reported a preferential flow in the horizontal direction using the laboratory cube method (Branham and Strack 2014; Lewis et al. 2012; Rosa and Larocque 2008). This pattern has been related to the presence of horizontal layers that limit vertical flow due to compression of the deep peat layers (Beckwith et al. 2003). However, in contrast to these studies and in agreement with our results, we did not find an increase in BD (related to compression) with depth (Table 2). At our site, the succession of peat layers does not follow a continuum, but alternates between layers with different properties. Couwenberg et al. (2022) described a similar pattern, which they called ‘zebra’ succession, in a peat bog in Tierra del Fuego. Furthermore, we have reported that the presence of roots increases vertical K (Figure 5C) precisely because of their mostly vertical arrangement. In this sense, Morris et al. (2019) concluded that the macropores created by the roots of ericaceous shrubs favour vertical permeability. Thus, the density and distribution of the rhizosphere are determining factors in the anisotropy and permeability of peat, which is often not analysed.

4.2 | Peat Classification and Regression Model

The proposed classification of peat layers allowed us to summarise some of the physical and hydraulic properties, as proposed by other authors (Liu and Lennartz 2019; Paat et al. 2022). The greatest differences were between the minerotrophic and ombrotrophic layers, so that the “transitional” was associated with the minerotrophic or ombrotrophic condition, depending on the property analysed (Table 3). The minerotrophic peat was characterised by a dark colour, abundance of fine remains of vascular vegetation and viscous texture when wet, whereas the ombrotrophic peat was light brown-yellow range in colour, porous, very fibrous and with little structure when wet. This visual characterisation agreed with our results, where the minerotrophic layers had lower KH and OM values, and higher BD and humification index values than the ombrotrophic layers. Surprisingly, although the trend was similar to the above, KV did not change significantly between peat categories. We associate this result with the higher variability of KV together with a higher proportion of samples with horizontal anisotropy in the ombrotrophic (higher anisotropy values). Reinforcing the idea of a “zebra” type succession, the types of categories did not differentiate with depth, that is we found ombrotrophic or minerotrophic layers both at the surface and at depth, indistinctly.

The regression models showed clear differences between the predictor variables for K in the vertical (KV) and horizontal (KH) directions. The categorical variable peat type (ombrotrophic/minerotrophic) and depth (as a quantitative variable) were significant predictors of $\log_{10}(K)$ in both models. However, KV was strongly influenced by root density (Table 4), whereas KH was mainly explained by bulk density (Table 5). Probably due to the high correlation found for OM and humification index with BD (Table 2), these two variables were not included in the model, as they were autocorrelated and did not provide any additional explanation. Studies by Morris et al. (2015, 2019) and Paat et al. (2022) have also obtained predictive models of K as a function of BD and depth in northern hemisphere peatlands. In these cases, the authors also found a dependence on the core or location of the profile (random effect), which was not significant in our case. In conclusion, the novelty of our study is the incorporation of root density in the evaluation of permeability and hydrology in the Fuegian peatlands, which will have a strong influence on the vertical flow of groundwater.

In recent years, quantitative studies on the hydrological functions of peatlands have increased, addressing processes such as groundwater recharge, water storage, surface runoff, and subsurface drainage of water and nutrients towards downstream areas (Lambert et al. 2022). However, these studies often simplify hydrological dynamics by using models that rely on a single value of hydraulic conductivity (K), generally expressed as a geometric mean (Branham and Strack 2014; Morris et al. 2022). Our results reinforce the notion of a strong spatial heterogeneity of this property, both in the horizontal dimension—driven by landscape microforms—and in the vertical dimension, associated with variations in bulk density, root density, and degree of decomposition (humification). The vertical variation is particularly relevant due to the fluctuations of the water level that occur during the growing season in Fuegian peatlands (Iseas et al. 2025). It is likely that during spring, following snowmelt

and under high water level conditions, surface runoff and lateral subsurface flow are enhanced, as the upper peat layers exhibit high permeability. Conversely, in summer, when the water level drops and reaches denser and less permeable peat horizons, lateral discharge becomes restricted and evaporative processes towards the atmosphere prevail.

4.3 | Evaluation of Field and Laboratory Methods

Comparisons between field and laboratory methods for hydraulic conductivity (K) are complex, mainly due to the assumptions inherent to each technique, the size of pore system or peat sample, and the hydric properties of the substrate (Reynolds et al. 2000; Pedescoll et al. 2011). Nevertheless, similar trends are generally expected. Previous studies have reported that field K values typically fall within the range of minimum and maximum laboratory K measurements (SurrIDGE et al. 2005). In contrast, Rosa and Larocque (2008) suggested that the variability associated with field and laboratory methods is smaller than the spatial variations caused by peat heterogeneity. In our study, field K values were lower than those obtained with the laboratory method. Despite these differences in absolute values, both methods displayed similar spatial patterns, with an acceptable correlation between them (Figure 4a). The hummock sites exhibited high K values throughout the profile, whereas the lawn sites had the lowest K values; this pattern was consistent for both methods.

Some of our field measurements, particularly in deeper layers, were atypically low despite following SurrIDGE et al. (2005) recommendations for piezometric tests. This may be explained by trapped gas bubbles near the inlet zone or by fine peat debris obstructing flow (Baird et al. 2004). Furthermore, water flow in the peat may be limited by an impermeable till sediment layer located near the surface (~1 m depth), which could partially account for the differences between field and laboratory K measurements. It is important to note, as a limitation, that although the cube method provides a simple way to characterise the variability of K and anisotropy, the sample size is typically small, not fully representative, and measured outside of its ecohydrological context. For this reason, we consider field determinations to be more representative of *in situ* conditions, as they reflect the natural hydrology of the peat with minimal disturbance (Baird et al. 2004; Hogan et al. 2006).

The MCM method (constant head) demonstrated high reproducibility, with a low coefficient of variation of the replicates ($mean = 7.7\%$) and produced K values comparable to those reported in previous studies (Branham and Strack 2014; Morris et al. 2019; Rosa and Larocque 2008). The two laboratory tests (falling head and constant head) showed a high correlation, with falling head determinations systematically lower than those obtained by the constant head method, similar to the results reported by Pedescoll et al. (2011, 2012). In our procedure, the first measurement was always performed using the constant head method, followed by the falling head test, which could potentially affect the saturation level of the samples and cause pore clogging due to particle rearrangement. However, Beckwith et al. (2003) demonstrated that the order of measurement does not significantly affect vertical or horizontal laboratory

determinations of K. Additionally, the use of deionised water with low electrical conductivity in our tests could influence colloid flocculation, reduce pore connectivity and thus decrease K values in successive measurements (Kettridge and Binley 2010). Although the constant head method produced lower experimental error in both orientations, falling head calculations were still acceptable. Rosa and Larocque (2008) indicated that an error of 16% is reliable, as it is negligible compared to the high variability between samples.

Therefore, we conclude that the falling head method, as implemented in the MCM procedure, is reliable due to its low experimental error, ease of implementation (since it does not require additional devices to maintain a constant head), and suitability for low-permeability samples (Bagarello et al. 2004; Chiasson 2012; Pedescoll et al. 2012).

5 | Conclusion

Measured values of saturated hydraulic conductivity offer valuable insights to enhance the understanding of the hydrology of the poorly studied southern peatlands. These values exhibit remarkable spatial variation with depth, likely influenced by the acrotelm/catotelm layers. However, our results suggest that although the permeability of the deep layers (catotelm) is low, there may be layers of poorly decomposed peat (ombrotrophic) with high permeability. Conductivity was significantly related to bulk density, humification, root density, and depth. These ecohydrological characteristics were adequately consistent with the classification of the layers according to the plant material remains (ombrotrophic, minerotrophic and transitional). Regression models indicated that vertical hydraulic conductivity variability is primarily explained by root density, depth, and layer type, whereas horizontal hydraulic conductivity is associated with bulk density, depth, and layer type. In addition, some degree of anisotropy was identified, but without any preferential flow.

Although it is difficult to make comparisons between techniques to obtain the “true” hydraulic conductivity of the substrate, the different methods were consistent. High and low permeability sites were well represented both in the field and in the laboratory. The falling head mode proposed for the modified cube method (MCM) correlated well with the constant head mode. Moreover, the low experimental error obtained is acceptable in the context of high variability of this property. Therefore, the MCM method with the falling head mode represents an interesting alternative since its application is simpler, more reliable, and especially recommended for low permeability samples.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Schematic representation of the Modified Cube Method (MCM) used to determine saturated hydraulic conductivity in the vertical (KV) and horizontal (KH) directions. Numbers indicate the sequence of steps: